Engineered Intelligence: Tracing Computational Design from Search to Evidence

A Retrospective Study of Representation, Search, and Assessment

Research draft prepared for Jepson Taylor · Human author approval pending · github.com/neverhuman/jarambit

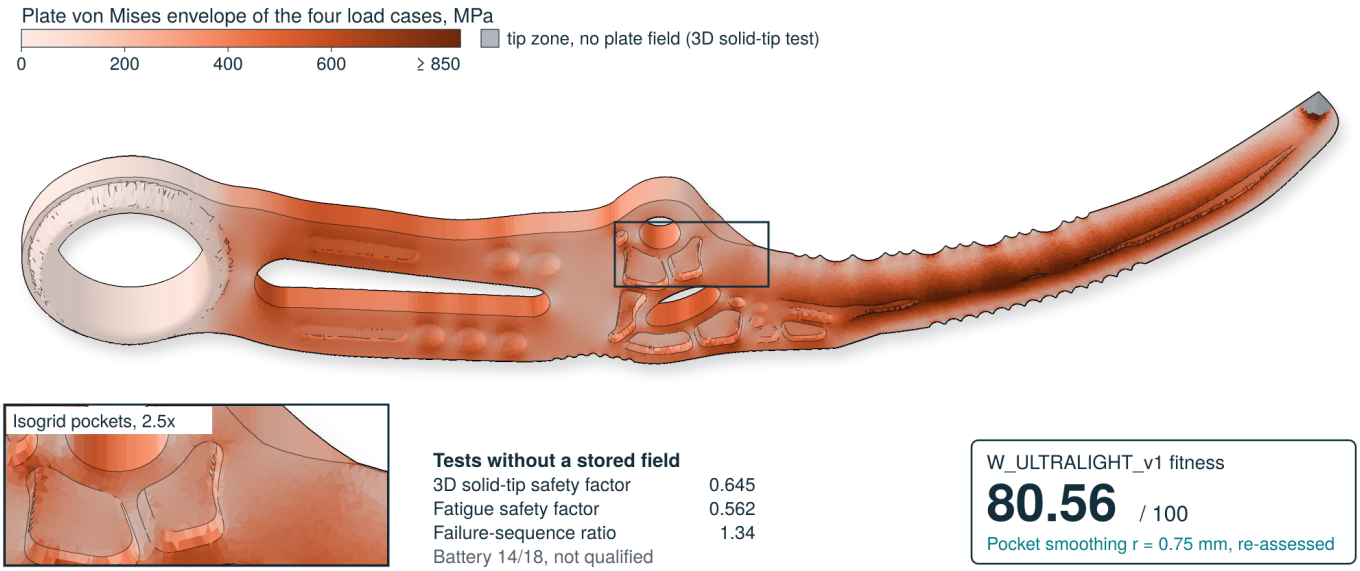


Figure 1 | Best evolved design by W_ULTRALIGHT_v1, `ec3i042_r01_p10_c56`, after surface-tension pocket smoothing ($r = 0.75$ mm; Section 6): re-assessed fitness $F = 80.561227$ of 100, against $F = 80.623796$ as exported (variant A). The smoother runs inside the CAD export of the same genome, so the smoothed design is a new STEP solid and every objective is measured again; it keeps the 5-point CAD/B-rep objective. 3D mass $83.56\,\mathrm{g}$, battery 14/18, not qualified. Same camera, light, $2.5\times$ isogrid-pocket inset and MPa scale as variant A. Colour is the plate-model von Mises envelope stored by this re-assessment, light for low and dark for high stress; grey is the 4 mm tip zone that only the 3D solid-tip test covers.

1,954,560
native birth records in 10 completed runs

1,761
comparable Stage-2 assessments, one canonical scorer

80.623796
best score as exported; a full re-assessment reproduces it

80.561227
the same genome after pocket smoothing, re-assessed in full

⊖ Not qualified
battery 14/18; not built; all results computational

Abstract

Automated design search can retain extensive records while leaving the evidence behind a selected result hard to interpret. We trace one computational result, the titanium hooked utility knife `ec3i042_r01_p10_c56` shown above, from its executable representation through recorded evolutionary search to its exported solid and assessment. Among 1,761 comparable Stage-2 assessments from 4 hosts, each recomputed by one canonical multi-objective score, it ranks first at 80.623796 of 100, and a complete re-assessment gives the same value. Surface-tension pocket smoothing inside the CAD export turns it into a new solid with clean pocket contours; its full re-assessment gives 80.561227. The records support five findings. Evidence coverage depends on the observation stage: 1,954,560 native birth records are not a large assessed population. A search rank is not an assessment outcome: the highest historical birth fails a recorded predicate that a lower-ranked one meets. A score is only as comparable as its scorer: an early evaluator overstated the canonical score of the same genomes by 11.97–13.12 points. Agreement between two solvers on a shared mesh does not admit extrapolated error estimates. A geometric post-process is a design change, judged by re-assessment rather than appearance. The contribution is an inspectable chain from representation to evidence, with incompatible scores and missing observations kept explicit. All results are computational; the design is not qualified and has not been built.

Index terms
Engineered Intelligence
computational design
evolutionary computation
executable genomes
CAD
quality diversity
finite elements
evidence provenance
reproducible populations

1 Introduction

Automated design search can now propose and score candidates by the hundred thousand. What remains hard is knowing what the retained records establish about the one design a paper puts on its first page: which representation produced it, which ranking rule preferred it, which assessment measured it, and which questions no record answers. This paper studies that question on one complete computational case.

We use *Engineered Intelligence* for an arrangement with an authored part and an automated part. People author a versioned representation and its decoder, a declared objective, and the tests that measure each exported artifact; an evolutionary search then proposes new candidates from the recorded scores. The term names this division of labour. It implies no particular optimizer and no learned model, and it does not remove human judgement: the problem, the representation and the tests remain authored.

The case is Jarambit, a titanium hooked utility knife with an integral finger ring. Its documented schema has 128 numeric coordinates, including conditional and discretized parameters, plus a categorical material field. The object is small but demanding: a mass objective interacts with lateral, ring-pull, torsion and fatigue requirements, and fine machined features (isogrid pockets, fullers, lattice and relief pockets) make both the exported CAD and its mechanics delicate.

The retained records are large and uneven. 10 completed native runs of 801 generations hold 1,954,560 birth records. A later week of island evolution on 4 hosts holds 77,915 Stage-1 screens and 1,761 comparable Stage-2 assessments of 1,719 distinct genomes. Every Stage-2 score in that census is recomputed by one canonical function, W_ULTRALIGHT_v1, from the design's own retained scorecard; stored leaderboard values are not used. The highest is `ec3i042_r01_p10_c56` at 80.623796 of 100. Because the top of a ranked list tends to be overstated [23], that design was re-assessed in full; it scores 80.623796. Its pocket-smoothed export is a new solid, also assessed in full; it scores 80.561227. Neither export is qualified, and neither has been built.

1.1 Research Question and Findings

We ask: *what can the retained links establish about a computational design result, and what can they not?* The records support five findings.

1) **Coverage depends on the observation stage.** Complete genome and score tables coexist with selective parentage, geometry and later-assessment records, so a large archive is not a large assessed population.
2) **A search rank is not an assessment outcome.** In historical Run 1 the highest-scoring birth, Individual 1095, fails that run's recorded certification predicate, which lower-ranked Individual 1543 meets; later all-requirement qualification is a separate assessment.
3) **A score is only as comparable as its scorer.** An early evaluator overstates the canonical score of the same genome (a stored 92.29 re-assesses at 79.166355). Its scorecards and those of a second evaluator matched to it, 142 in all, are excluded, and every remaining score is recomputed.
4) **Solver agreement and error estimation answer different questions.** Two finite-element implementations agree within 0.0448 % on a shared solid mesh, while the plate mesh study refuses its extrapolated error estimates.
5) **A geometric post-process is a design change.** Surface-tension pocket smoothing preserves the pockets' removed volume to within 0.014 %, yet the same setting moved the score by +0.071015 on the design used to choose it and by −0.062569 on the current best, so it is judged by a complete re-assessment, design by design.

1.2 Contributions and Scope

The paper contributes a source-linked account of one result, from representation and decoder through search records and canonical score to exported solid and assessment; a census of every comparable Stage-2 assessment under one scorer, with the exclusions and re-assessments that make it comparable; a volume-preserving smoothing step inside the CAD export, evaluated by the full assessment rather than by appearance; and a map of what each retained observation supports, including refused error estimates and a withdrawn buckling inference. Section 3 defines the records and counting units, Sections 4–7 the representation and inheritance, Section 9 the census, and Section 11 the numerical evidence.

> **Scope of the claims**
>
> All results are computational. No evolved design was fabricated or physically tested, and the best design is not qualified. Designs are ranked only by the canonical W_ULTRALIGHT_v1 multi-objective score; legacy requirement checks are measured and reported but do not veto selection. No controlled comparison with other optimizers or with expert designs was run, so no search advantage is claimed.

2 Background and Related Research

2.1 Computational Design and Representation

Computational design already includes mature approaches to changing structural form. Sigmund and Maute review density, level-set, evolutionary, and other topology-optimization approaches [13]. Jarambit's documented native schema instead varies an application-specific parameterization under a fixed decoder. Its coordinate count is not evidence of an unrestricted topology search or a newly discovered representation language.

DeepCAD learns CAD-operation sequences for autoencoding and generation [2]; Text2CAD generates parametric CAD sequences from text and reports visual, parametric, and geometric evaluations [3]. These representation choices differ from Jarambit's numeric schema. None supplies a common mechanical benchmark for the present study.

Evolutionary engineering also predates language-model assistance. The cited 2003 ST5 report describes evolved antenna representations, fabricated prototypes, and comparison with a hand-designed antenna; flight qualification was still pending in that report [4]. MAP-Elites retains representatives across user-defined behavioral dimensions [5]. Such archive selection differs from retaining all observed births, and archive occupancy alone does not measure population diversity. Jarambit's cohort-specific archive and score definitions therefore remain part of its reported context.

2.2 Research Provenance and Artifact Access

The closest context for this study's evidence links is research provenance. PROV-DM distinguishes entities, activities, derivations, responsible agents, and collections [14]. These concepts help explain why a candidate record, a search activity, an exported geometry, and a reporting derivative require distinct identities. Jarambit's retained links are compared on these dimensions; no PROV conformance or general interchange implementation is established here.

Implemented provenance systems provide a closer comparison than metadata principles alone. AiiDA stores calculations, workflows, inputs, and outputs in a traversable provenance graph and supports automated computational workflows [18]. Workflow Run RO-Crate packages workflow-execution provenance with associated inputs, outputs, and code across execution systems [19]. These systems already address provenance capture and exchange.

Jarambit contributes a retrospective case analysis at the boundary between records and published engineering claims: which candidate and observation stage a quantity describes, which denominator a count uses, and which evidence remains unavailable. The present study does not benchmark its packaging against AiiDA or RO-Crate, or establish superior capture, interoperability, or reader comprehension.

The FAIR principles address discovery, access, interoperability, and reuse of research objects [15]. Software citation principles additionally distinguish persistent identification, access, and specificity of the software version used [16]. A hash identifies bytes; meaningful reconstruction also requires retrievable artifacts and interpretation of their types, revisions, and relationships. This study's contribution is a concrete retrospective account of those relationships and gaps, rather than a new general provenance model.

We cite the inspected Jarambit publication baseline as a source snapshot [1]. It is distinct from the cohort producers and from the final manuscript release accession (Section 12). A methods publication, a source snapshot, an executed binary, and a generated figure likewise denote different objects.

2.3 Numerical Credibility and Visual Interpretation

NASA's models-and-simulations standard supplies context for separating credibility questions; compliance is not claimed [7]. The scikit-fem and DOLFINx methods references identify finite-element software roles, while execution provenance must identify the versions and model inputs actually used [8], [9]. Selected observable agreement on a shared mesh and admissibility of a grid-refinement estimate answer different questions (Section 11).

Quantitative views also need explicit units, transformations, masks, and legends. Crameri *et al.* explain how uneven color gradients can introduce visual distinctions absent from the data [17]. A percentile summary therefore requires its own interpretation and cannot inherit the physical units of its component fields. The paper's spatial views are reporting devices, not a separately validated visualization method.

3 Study Design and Retained Evidence

This is a retrospective study of selected retained records, rather than a prospective comparison of optimizers. We ask what those records establish about representation, recorded search preference and later numerical assessment. The three cohorts have different scoring and retention policies; the separately assessed exports supply additional candidate-specific observations, rather than downstream measurements for every search birth. Figure 2 separates these retained evidence paths. Individual 2000 is a stable publication alias for a retained export; the integer does not state its rank, birth sequence or generation. Exact certificate and genome hashes link it to a retained proposal, while its per-child parent identifiers remain unrecorded.

A *birth* is a recorded candidate-creation event. A *membership* is an appearance in an island slot, including retention or migration of an existing individual. An *archive representative* is a selected record under its archive's objective and descriptor context. An *assessed export* has a specifically bound artifact/assessment bundle. These are different counting units: one genome can occur in several births or memberships, and a certificate identity alone does not establish that its CAD or assessment artifact is retained.

Sources H and P are the historical reconciliation and proposal/screen join accessions; N is the completed generation-800 population accession. E is the automatic manifest plus original field/assessment/verification contracts; C is the controlled-export scorecard accession. Availability is local to these accessions. The native population tables retain all 128 numeric coordinates and material, but their size does not establish complete parentage, per-birth exported CAD, later solid assessment or independent physical replication. Their accession records an untracked producer state rather than a clean executed-source binding; five of the ten completed-run manifests have no process-configuration hash. The Supplement distinguishes that accession statement from the separately retained baseline and local-patch hashes.

3.1 Run Identities and Capture Overlap

The native inventory contains 18 current campaign directories: ten completed generation-800 runs and eight others. An earlier staged snapshot has eleven summary files, of which four are earlier captures of runs later included among the ten completed runs (public Runs 6–9). These are overlapping views of original run identities, not 21 independent runs. Nine aborted crescent attempts are a separate accession; seven retain 216 summary rows in total and two have no retained CSV rows. Summary rows describe generation/island aggregates, not new births. Capture/receipt times are kept separate from unverified historical process start/stop times.

3.2 Selection and Interpretation

The selection registry designates Individual 2000 as the illustrated assessed export because its retained bundle connects export geometry, plate fields, mesh-study observations, solid comparison and scorecard. Its recorded human editorial-authority category does not independently identify a named chooser. This illustration role confers neither championship nor

Table 1 | Context by representation, search, evidence, and validation scope. Rows describe distinct research purposes rather than a performance ranking.

Context	Representation	Search or transformation	Evidence emphasized	Validation scope
Topology optimization	Varies by formulation	Numerical design updates	Structural objectives and constraints	Method- and model-specific
DeepCAD / Text2CAD	CAD-operation sequences	Learned generation	Reconstruction or generated CAD	Reported geometric/parametric evaluations
Provenance systems	Workflow objects and relations	AiiDA capture; RO-Crate packaging	Execution history, artifacts, and derivation	Provenance/exchange scope, not physical validation
This study	Fixed CAD schema and candidate identities	Retained evolutionary and assessment records	Coverage, ancestry, scores, and selected numerical observations	Retrospective computational evidence

Table 2 | Coverage of the admitted study records. A: available; P: partial/selected; NR: not retained in the named accession; U: coverage not established. Physical observations are not reported in this study (NPub); this does not infer that no experiment ever occurred. Each row's source key applies to all its cells; the Supplement supplies exact paths, hashes and qualifications.

Records (source)	**Denominator**	**Genes/ material**	**Parentage**	**Exported geometry**	**Spatial fields**	**Later assessment**	**Physical**
Historical genealogy (H)	1984 births; 2176 memberships	A	A within run	NR in cited accession	NR in cited accession	P: selected strict records	NPub
Proposal/screen campaign (P)	3649 proposals; 2259 screens	A in proposals	NR per-child IDs	P: selected exports	P	P: selected summaries/bundles	NPub
Ten completed native runs (N)	1,954,560 births	A	U beyond admitted tables	P: separate STL accession	NR per birth	NR per birth	NPub
Automatic assessed export (E)	Individual 2000; one assessed	A	NR per-child parents	A	A plate; NR full solid	A saved bundle	NPub
Controlled exports (C)	Individuals 2001/2007, separately scoped	A	U birth link	A	P	A saved scorecards	NPub

qualification. Table 6 separates it from source-bound stored score leaders and other assessed records. The historical leader illustrates a later-assessment disagreement; neither example is a random sample from the large native archive. This reporting reconstruction does not invent scores for missing observations or turn them into a measured qualification fraction. Historical weighted-score defaults are preserved and disclosed separately in Section 8.

4 Representation and Candidate Identity

4.1 A Gene Is a Semantic Contract

A gene requires an explicit coordinate index, physical or discrete type, unit, domain, coupling law, and decoder revision. A serialized floating-point token alone cannot distinguish a millimeter displacement from an angle, a feature-count selector, or an architecture switch. Those semantics determine what a stored genome means when it is decoded.

The documented native schema stores 128 numeric coordinates and one categorical material field. Some numeric coordinates express continuous parameters; others are interpreted through thresholds or discretization. The nominal coordinate count is therefore neither the number of independently active parameters nor a measure of geometric freedom. Field roles are summarized below; exact domains and decoding rules require the producer revision attached to each cohort. Historical-producer details are retained in Supplement 1, without attributing them to every executed workflow.

Table 3 | Functional groups in the documented numeric schema.

Group	Role and stored type	Conditional interpretation
Silhouette	Numeric normal offsets around a reference outline	Some stations can be pinned by the producer's bounds.
Planform and features	Dimensions, angles, ratios, and feature selectors	A stored numeric feature setting need not activate a feature.
Internal machining	Pocket, window, and perforation parameters	Feature counts and zone switches use discrete decoding rules.
Continuous-rail layer	Numeric architecture switch, dimensions, and thickness knots	The switch selects the thickness architecture; only the selected knot count contributes.

4.2 Coupling and Conditional Activity

Clamping, thresholding, and coupled bounds distinguish stored coordinates from the geometry they produce. In the bound historical producer, the continuous-rail switch is thresholded, the active knot count is floored and bounded, and enabling the layer disables several legacy features. When it is disabled, knot coordinates are set to stock thickness during clamping. The decoder and its configuration thus belong to the interpretation of a record, rather than being incidental implementation details. A numeric change can be inactive, discretized away, or constrained by another coordinate; changed-coordinate counts alone do not establish geometric change.

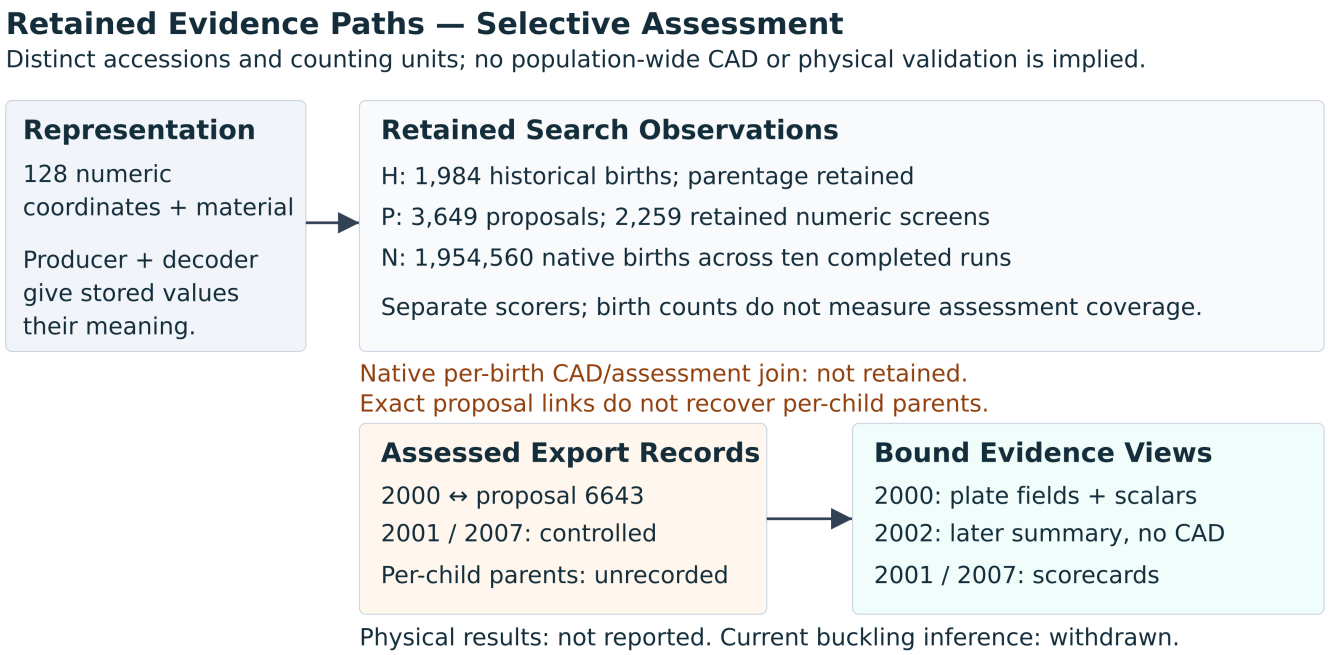


Figure 2 | Distinct retained evidence paths. Search observations and selected assessed exports have separate denominators and source bindings; a complete per-birth native CAD/assessment join is not retained. An arrow connects an assessed export only to its own retained evidence. Physical observations are not reported in this study.

4.3 Birth, Genome, Membership, and Export

A birth identifies a recorded generation event; a genome serialization identifies stored coordinate and material values; a membership identifies a population-slot occurrence; and an assessed export identifies its own geometry and assessment bundle. These objects can share an individual identifier without sharing a denominator. The historical cohort, for example, retains repeated genome serializations and repeated appearances of earlier births. A display mesh, a plate mesh, and a solid submodel likewise represent distinct discretizations, each requiring its own source binding (Section 12).

The inheritance ledger counts changed numeric coordinates by exact numeric inequality on parsed CSV values. Its raw-coordinate L_2 norms mix units and discrete semantics. They are audit summaries of stored coordinates, not physical distances between shapes; the detailed edge ledger is retained in Supplement 1.

5 Observed Morphology and Feature Provenance

The best evolved design by W_ULTRALIGHT_v1, ec3i 042_r01_p10_c56 (F = 80.623796 as exported, 80.561227 after pocket smoothing, Figure 1), makes the representation tangible: its retained Stage-2 bundle, re-assessed with the current evaluator, connects a geometric object with its genome, exported STEP and STL, scorecard and solid-tip observations, and Figure 3 shows that object in relief. The earlier illustrated example, Individual 2000, still supplies the retained plate response fields analysed in Sections 8–11. Three questions organize the interpretation: which feature families the representation makes available, which features an identified artifact actually contains, and what evidence establishes their consequences.

5.1 An Authored Feature Vocabulary

The documented representation provides silhouette and opening controls alongside longitudinal channels, face relief, patterned pockets, perforations and ring grooves. The producer's genome and phenotype sources define these families explicitly, including a blade fuller, triangular isogrid pockets, lattice features and ligament flutes; the reference source separately defines a ring-face groove[1]. These are authored parts of the vocabulary available to the decoder, rather than feature families whose invention is demonstrated by the illustrated export.

This vocabulary gives ordinary names to otherwise abstract coordinates. A fuller names a channel family; a pocket names a recessed region; a perforation names a through-opening. Material remaining between openings can be described as a bridge without assigning it a particular structural cross-section. Conditional switches, coupled bounds and construction rules determine how stored settings become geometry (Section 4). The available vocabulary and an individual feature's realization are therefore distinct observations.

5.2 Occurrence Requires an Identified Geometry

An occurrence claim belongs to a particular export and decoder context. The appropriate evidence is the retained geometry with its candidate identity and source binding. A nonzero gene setting alone does not establish the realized shape. Likewise, the boundaries in a response-colored image identify changes in a displayed field; they are not measurements of a geometric cross-section.

Neutral geometry and response views should consequently be read together, with a consistent orientation and an explicit account of the pictured region. Additional local anatomy callouts require a bound geometry asset and crop transformation.

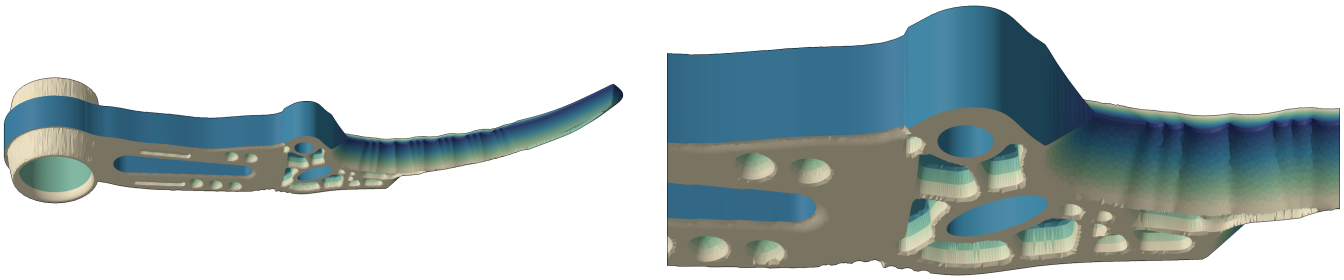


Figure 3 | The best evolved design, ec3i042_r01_p10_c56, after surface-tension pocket smoothing at $r = 0.75\,\mathrm{mm}$ (Section 6; the as-exported relief is level L0 of the smoothing ladder), in relief: whole design (left) and a close-up of the guard and pivot zone (right) from one oblique camera under a low raking light, with the thickness direction exaggerated $2.5\times$ and faces coloured by height above the mid-plane (dark blue at the mid-plane to cream at the $2.62\,\mathrm{mm}$ stock half-thickness; the finger ring saturates). Visible are the finger ring, the handle slot, two-sided isogrid pockets around the pivot and guard (full-thickness ribs between floors $1.35\,\mathrm{mm}$ deep on each face, an I-beam-like section), a pierced-fuller window, rows of relief pockets along the handle, the ground blade bevels and the edge serrations. Its ancestor w406t3_r01_p10_c16 (80.299390) has the same pocket layers; 19 of 129 genes differ, mostly slightly. Same renderer and settings as level L0 of the smoothing ladder (docs/paper/smoothing-ladder.pdf).

Until those bindings are supplied, descriptive names remain pointers for inspection rather than authenticated measurements of the highlighted regions. A content match can identify a recorded occurrence; a nearby genome does not establish a parent or the history of an individual feature.

5.3 Utility Is a Separate Engineering Claim

The saved assessments concern the illustrated candidate under their stated models and loads. They do not isolate the contribution of one channel, pocket or opening. This section therefore describes the representation and its inspectable artifacts without assigning feature-specific mass savings, efficiency, optimality or manufacturing qualification. Its value is explanatory: readers can connect the numeric representation to a concrete object while keeping feature availability, observed occurrence and demonstrated utility separate.

6 Surface-Tension Pocket Smoothing

Why the walls were chipped. The exported solid is a height field over a fixed planform: faces $z = \pm t(x, y)/2$, with side walls on the outline and on every through-hole. Every weight-saving feature (isogrid pockets, lattice dimples, fullers, flutes, relief) is a removal depth $D = t_{\mathrm{base}} - t$. Unsmoothed, that depth reaches the CAD through the 0.25 mm thickness raster by nearest-cell lookup, so most pocket walls are vertical steps along a staircase. A constrained Delaunay triangulation (CDT) of about 0.7 mm triangles cuts across those steps, so the walls render as chipped (Figure 4b, left). Mesh smoothing (docs/paper/smoothing-ladder.pdf) cannot remove this: it pins every vertex on an edge sharper than 30°, which is where the chips are.

Method. We smooth the removal depth itself, before meshing, as a liquid would. Each pocket is a basin: a connected component of $D > 0.02\,\mathrm{mm}$ (a watershed), merged across gaps narrower than $r/2$. Inside its basin the depth evolves for time $T = r^2/2$ by level-set curvature motion [21] with a per-basin pressure,

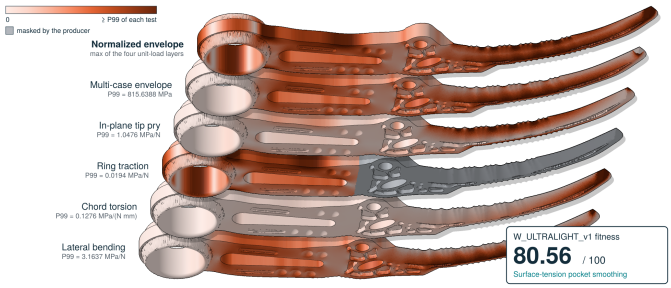
$$D_t = |\nabla D|\,\nabla\cdot\frac{\nabla D}{|\nabla D|} + \lambda_b\,|\nabla D|, \qquad (1)$$

where λ_b keeps the removed volume of basin b constant (the volume of oil in the pit). Every depth contour then shortens like a two-dimensional droplet of fixed area under line tension [22]: chips and spikes, which have the highest curvature, go first; straight rib edges stay straight and corners round to a radius of about r; and walls keep their steepness (rib widths were inspected in renders, not measured). No basin may end deeper than its own deepest original point, so the volume constraint widens a thin groove instead of drilling it toward the minimum web. The planform, through-holes, grind faces, bore collar, raised features, brand pad and a 1 mm rim of unpocketed material along the outline and holes never move.

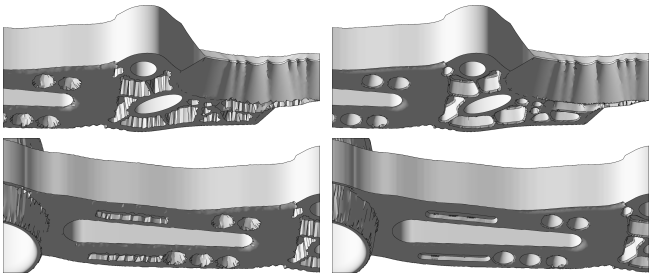
The flow runs on a 0.125 mm grid (1.7–2.9 s per design); the exporter reads $t = t_{\mathrm{base}} - D$ by bilinear interpolation and adds CDT vertices in three rows along every wall (depth contours at 8, 50 and 92 % of the local wall height) and in pairs straddling the grind boundary, so triangle edges follow each wall instead of cutting across it. Equation (1) is the $a = 0$ case; $a = 1$, a liquid-film flow, is compared below. The switch is JARAMBIT_POCKET_SMOOTH (crates/crescent/src/pocket_smooth.rs); on selects $r = 0.75\,\mathrm{mm}$ and $a = 0$. It is off by default, and an enabled setting is part of the certificate identity.

Settings and fitness. We chose the setting on the previous best design, w406t3_r01_p10_c16, exporting it at $r = 0.5$, 0.75, 1.0, 1.5 and 2.0 mm with $a = 0$, and at $r = 0.75\,\mathrm{mm}$ with $a = 1$. Every export was re-assessed with the complete Stage-2 evaluator, so all twenty objectives, including the STEP solid, are measured on the smoothed geometry; mesh smoothing [20], by contrast, loses the 5-point CAD/B-rep objective (75.45–75.68on the ladder). Table 4 gives the results together with an unsmoothed re-run of the same evaluator, which reproduces the retained 80.299390 exactly. The current best design was then assessed at the chosen setting only.

Results. On w406t3_r01_p10_c16, smoothing leaves the engineering responses essentially where they were. The removed volume of every basin is preserved to rounding, and the exported volume changes by $-3.19\,\mathrm{mm}^3$ (−0.017 %) at $r = 0.75\,\mathrm{mm}$. Mass stays at 83.19–83.26 g against 83.2231 g as exported; the lateral capacity rises by 0.2–0.8 N; the ring, body-twist and fatigue margins move by less than 3 %; the



(a) Retained plate responses of the smoothed design



(b) Raw export (left) and $r = 0.75$ mm (right): guard, handle

Figure 4 | Surface-tension pocket smoothing of the best design, ec3i042_r01_p10_c56, at $r = 0.75$ mm ($F = 80.561227$). (a) The five retained plate responses of the smoothed design (bottom to top: unit lateral bending, chord torsion, ring traction, in-plane tip pry and the multi-case envelope), each normalized by its own area-weighted 99th percentile, with their normalized envelope on top; colour is a within-test rank, not a safety factor. (b) Guard (isogrid pockets) and handle (flutes, lattice dimples), raw and smoothed, under the ladder's guard camera with $2.5\times$ thickness; the outline, holes and grinds are unchanged, and the pockets shown keep their volume, depth and rib widths.

Table 4 | Complete Stage-2 re-assessment after pocket smoothing: the settings study on the previous best design, w406t3_r01_p10_c16 (top), and the current best design, ec3i042_r01_p10_c56, at the chosen setting (bottom; ΔF against its own as-exported 80.623796, battery 13/18). F is W_ULTRALIGHT_v1 out of 100; the tip columns are the 3D P2 solid-tip safety factor at 1.82 N m (nodal maximum and average); lateral is the hunter capacity at certification and strict meshes.

Setting	F	ΔF	Mass (g)	Tip SF	Lateral (N)	Battery
As exported (re-run)	80.299390	–	83.2231	0.6823 / 0.7569	181.5 / 182.2	14/18
$r = 0.5$, $a = 0$	80.164307	−0.135083	83.2178	0.6353 / 0.6645	181.7 / 182.4	15/18
$r = 0.75$, $a = 0$ (chosen)	**80.370405**	**+0.071015**	83.2089	0.7024 / 0.7511	181.8 / 182.5	14/18
$r = 1.0$, $a = 0$	80.308595	+0.009205	83.2070	0.6811 / 0.7237	181.9 / 182.5	14/18
$r = 1.5$, $a = 0$	80.158645	−0.140745	83.1869	0.6275 / 0.6751	182.2 / 182.7	14/18
$r = 2.0$, $a = 0$	78.275029	−2.024361	83.2001	unmeasured	182.3 / 182.8	13/18
$r = 0.75$, $a = 1$	80.161219	−0.138171	83.2629	0.6349 / 0.6752	182.2 / 182.7	14/18
Current best, $r = 0.75$	**80.561227**	**−0.062569**	83.5582	0.6450 / 0.6818	201.1 / 199.9	14/18

failure-sequence ratio rises from 1.29 to 1.45–1.62; and the CAD/B-rep objective keeps its five points in every case.

What moves F is the 3D solid-tip submodel. Its nodal-maximum safety factor changes non-monotonically with r (Table 4). At $r = 2.0$ mm the tip solve produced no observation, which the evaluator scores as zero. We read these changes as the sensitivity of a nodal maximum near the fuller end to the local triangulation, not as a structural trend (the two solvers agree within 0.08 %). The setting $r = 0.75$ mm, chosen visually, is also the best measured on that design.

Larger r merges small pockets into rounded blobs, and the film variant ($a = 1$) spreads depth into the ribs; neither improves F. The chosen export has about 40 % more triangles (STEP 86.7 against 53.4 MB) and a larger tip mesh (177,984 against 91,008 tetrahedra).

Current best design. At the chosen setting, the design of Figure 1 scores $F = 80.561227$, 0.062569 below as exported: lateral capacity, failure sequence and fatigue gain 0.069 points, and the solid-tip safety factor costs 0.131 (setting not re-tuned). One 0.22 mm^3 sliver of its 15 basins fills in; the rest keep their volume.

Scope. The smoothing changes interior pocket geometry only and adds no qualification or physical observation. The Individual 2000 plate fields of Section 8 are retained evidence of a different design and are not smoothed. Receipts, scorecards and every exploratory setting are in experiments/pocket-surface-tension-smoothing/.

7 Recorded Search and Inheritance

7.1 Four Collections with Different Meanings

Births, population memberships, archive representatives, and assessed exports answer different questions. Births record proposed individuals and their observations. Memberships record occupied island slots, including retained individuals. Quality-diversity archives retain representatives under a run's descriptor and score rules. Later assessments concern selectively admitted or separately exported candidates. Evidence coverage and denominators for these collections are reported with the cohort-specific results (Section 9).

For the historical cohort, the recorded membership stage is *after contextual rescoring and before ring migration*. The bound producer emits birth or elite events and summary observations before migration. Migration subsequently copies selected IDs to destination islands; the overwritten destination IDs are not retained by those events. Calling the earlier snapshot a generation-end post-migration population would therefore overstate what can be reconstructed. Elite events retain membership but omit refreshed contextual fitness, so an individual's stored birth score is not substituted for an unrecorded later score.

7.2 Variation in the Historical Producer

The historical producer uses four island populations, shared novelty memory, a quality-diversity archive, elite retention, tournament selection, and periodic ring migration. Parents can be selected from the archive or current population; recorded parent IDs and operation labels identify the observed birth mechanism. These are source-bound historical semantics, rather than an assertion that every retained cohort executed an identical binary.

Crossover inherits the silhouette's 40 control points as one block and chooses the remaining numeric coordinates and material category independently between parents, then clamps the result. Mutation selects among six modes: sparse coordinate perturbation, a smooth silhouette bump, a thickness-and-section operator, a lattice operator, a fuller/texture/feature operator, and a machining-or-rail operator. Mutation scale depends on island and configuration; the exact parameter branches and the

coarse/fine distinction are retained in Supplement 1. This is variation within a fixed representation and decoder.

7.3 Ancestor Reuse in the Retained Ledger

Figure 5 uses the current generated inheritance ledger's three-hop view. Individual 1095 has recorded parents 413 and 526, and 526 itself has parents 413 and 392. Individual 413 is therefore reused along the paths 413 → 1095 and 413 → 526 → 1095. The dashed 174 → 299 link is a recorded relationship retained as induced closure, distinct from the three-hop edges. The ledger preserves exact coordinate differences and parent ordinals; the main figure emphasizes relationships rather than repeating their raw mixed-unit magnitudes.

This ancestry illustrates a reconstruction capability. Its focal birth's high stored search score and later assessment disagreement are separate observations, developed in Section 9; neither ancestry nor score alone establishes physical performance.

8 Score Definitions and Comparability

A search preference, a stored reporting vector, a shortfall score and a mechanical safety factor answer different questions. Their names, directions, observation stages and missing-input rules must accompany their values. Table 5 keeps the score regimes separate; a shared word such as "fitness" does not create a common performance axis.

8.1 A Retained Weighted Reporting Vector

The principal reporting definition is

$$F = \sum_{i=1}^{20} w_i s_i, \qquad s_i \in [0,1], \quad \sum_i w_i = 100. \tag{2}$$

Its priorities are mass/compactness (30%), form/comfort proxies (25%), exported geometry (15%) and response/verification (30%). Supplement Table 12 shows sums of *stored weighted contributions* for Individual 2002. The Supplement retains all twenty normalized values, weights and contributions. These are reported observations from its summary, not a recalculation under the present scorer and not measurements transferred from Individual 2000.

The saved scalar is $F = 74.207299$, with reported raster mass 101.139 g and solid-mass scalar 101.1384 g. Its exact genome and certificate link to a retained proposal and screen; a separate leaderboard retains a later-assessment summary with qualified=false. The underlying exported geometry and full scorecard files are not retained at its referenced paths in the checked repository tree. The screen's earlier "pending independent top-k assessment" label is not that later summary's outcome. Individual 2000 supplies the illustrated assessed geometry and response fields.

A versioned twenty-term recipe is recoverable, but its exact historical execution and the curves' point stages remain unresolved. Several saved inputs match that recipe's hard-coded fallbacks for bore radius, edge/point profiles and overload proxies. The four whole terms containing those inputs contribute 15.682353 points; this is not a claim that every point in those terms is an imputed measurement. Some absent-key Boolean branches also receive favorable defaults. The Supplement distinguishes these default-compatible inputs and historical bookkeeping from new reporting-layer imputation. The original vector is preserved, without rescoring.

For the assessed Individual 2000, the actual failed-branch score is $W_{\mathrm{CAD}} = -5.570408$. The same scorecard stores unpenalized quality 1.456983; that is the hypothetical pass-branch value, not an additional achieved score. The Supplement gives the branch equations, input names and shortfall arithmetic. Those scores and G_1 remain separate from F and native f.

8.2 Context and Display Direction

Native fitness includes novelty computed against contemporaneous population/archive descriptors. A value can therefore change when its context changes, even if the genome does not. Birth scores and later membership observations also have distinct stages. Cross-generation or cross-run maxima describe retained contextual preference; they do not isolate a physical improvement. The historical and newer native implementations/configurations require separate provenance even when both call their scalar f.

A display loss $L = -f$, $100 - F$, or $-W_{\mathrm{CAD}}$ reverses numerical order. Minimizing it preserves the original preference for a larger native score. It does not make those different profiles comparable. A cumulative maximum is monotone by construction, so its corresponding loss is nonincreasing by construction; that fact alone is not evidence of continuing physical improvement.

The missing-aware response display (the roster of retained channels, the per-cell percentile index and its coverage) is defined in Supplement Section 7.1.

Figure 6 gives source-coordinate views of the envelope and three physical-unit responses; the fourth eligible response, in-plane tip pry, appears in Supplement Figure 19C. Its saliency policy selects a concentrated upper-value region only when whole-field coverage, compactness and scale-gain requirements are met; otherwise it retains full context. B/D display a wider context camera spanning at least 30% of the whole object's horizontal extent, while keeping the selected region's statistics separate. An upper-value cutoff is a fraction of the original eligible value range, not a count or area quantile. Gray denotes producer-masked cells, not measured zero stress. Raw torsion retains its tip-patch peak, distinct from the post-exclusion peak used by selected capacity calculations.

9 Findings from the Retained Records

9.1 Large Record Coverage Does Not Imply Rich Assessment Coverage

The ten completed native runs (Runs 2–11) retain 1,954,560 evaluated birth records, 195,456 per run across generations 0–800. The admission receipts separately count 32,040 generation/island summary rows. All birth rows preserve the numeric coordinates and material, but no per-birth later CAD/solid assessment is retained for this population. The largest archive therefore supplies search-record coverage rather than a qualification rate.

Supplement Figure 16 keeps the native population separate from the historical 1,984 births and the intermediate campaign's 2,259 numeric screens. The 1,390 proposals lacking screens have no invented score. Within the native archive, 754,236

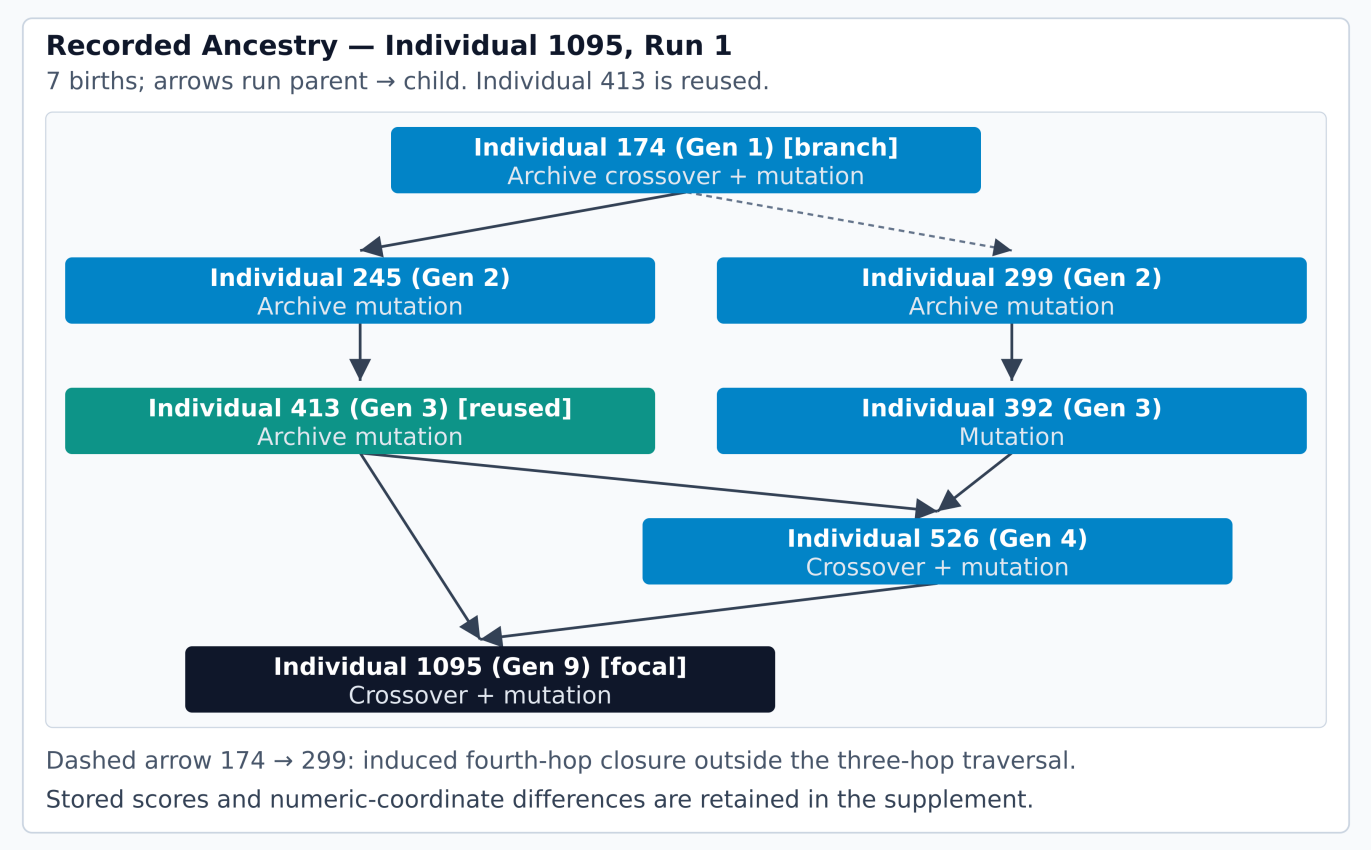


Figure 5 | Shared ancestry in the historical search record around Individual 1095. Solid edges show recorded parent-to-child links within three parent hops; the dashed edge $174 \to 299$ closes the induced subgraph but lies four hops from the focal individual along that path. Individual 413 contributes through both a direct and an indirect path. The genealogy belongs to this historical cohort.

Table 5 | Score dictionary. Scores are dimensionless; physical units remain attached to their inputs. The reporting reconstruction invents no missing scores; historical scorer defaults are disclosed below.

Quantity / stage	Direction / scale	Meaning and normalization	Comparison / missingness
Native f; historical and completed native search	Higher preferred; conditional negative branches, no common $[0, 1]$ scale	Mechanical/mass terms and contextual novelty after legality/feasibility branches	Bound profile, configuration and stage required; novelty changes with context; negative legal scores occur in completed native runs
Screening deficit; intermediate proposals	Lower shortfall preferred; nonnegative stored sum	Stage-1 screening quantity; not the later 20-term reporting vector	Only observed screens have numeric values; 1,390 proposals lack screens
Weighted F; retained reporting curves	Higher preferred; $[0, 100]$	Twenty saved normalized terms with weights totaling 100	Compatible versioned recipe recovered; exact executed producer and point-stage comparability unresolved; some inputs are default-compatible
W_{CAD}; 16-gate CAD scorecard	Higher preferred; conditional positive/negative branches	Pass-branch quality versus failed-branch normalized-shortfall penalty	Same assessment definition/inputs required; a favorable subscore is not all-gate success
G_1; assessment composite	Higher preferred; dimensionless	Weighted solid-tip factor, strength/mass, shape, mass penalty and sequence gap	Missing required inputs produce incomplete status; a screen-tip substitute is marked provisional; does not override gates

births (38.59%) are search-legal; 610,051 of those have negative contextual fitness. Score sign consequently cannot stand in for legality. Per-run legal counts range from 70,278 to 80,132, with mean 75,424 and sample standard deviation 2937 births. Run-maximum stored fitness ranges from 1.1089 to 1.1440 (mean 1.1250, sample standard deviation 0.01285). These are descriptive summaries of ten selected runs; related births do not provide millions of independent replicates.

Supplement Figure 6 adds directed tail detail: empirical Q90 through the actual retained maximum for native and historical higher-is-better fitness, and the actual minimum through Q10 for lower-is-better screening deficit. Cutoffs use $\lfloor p(n-1) \rfloor$ and include all boundary and favorable-extremum ties. Each panel reports its original-unit bounds and exact below/inside/above

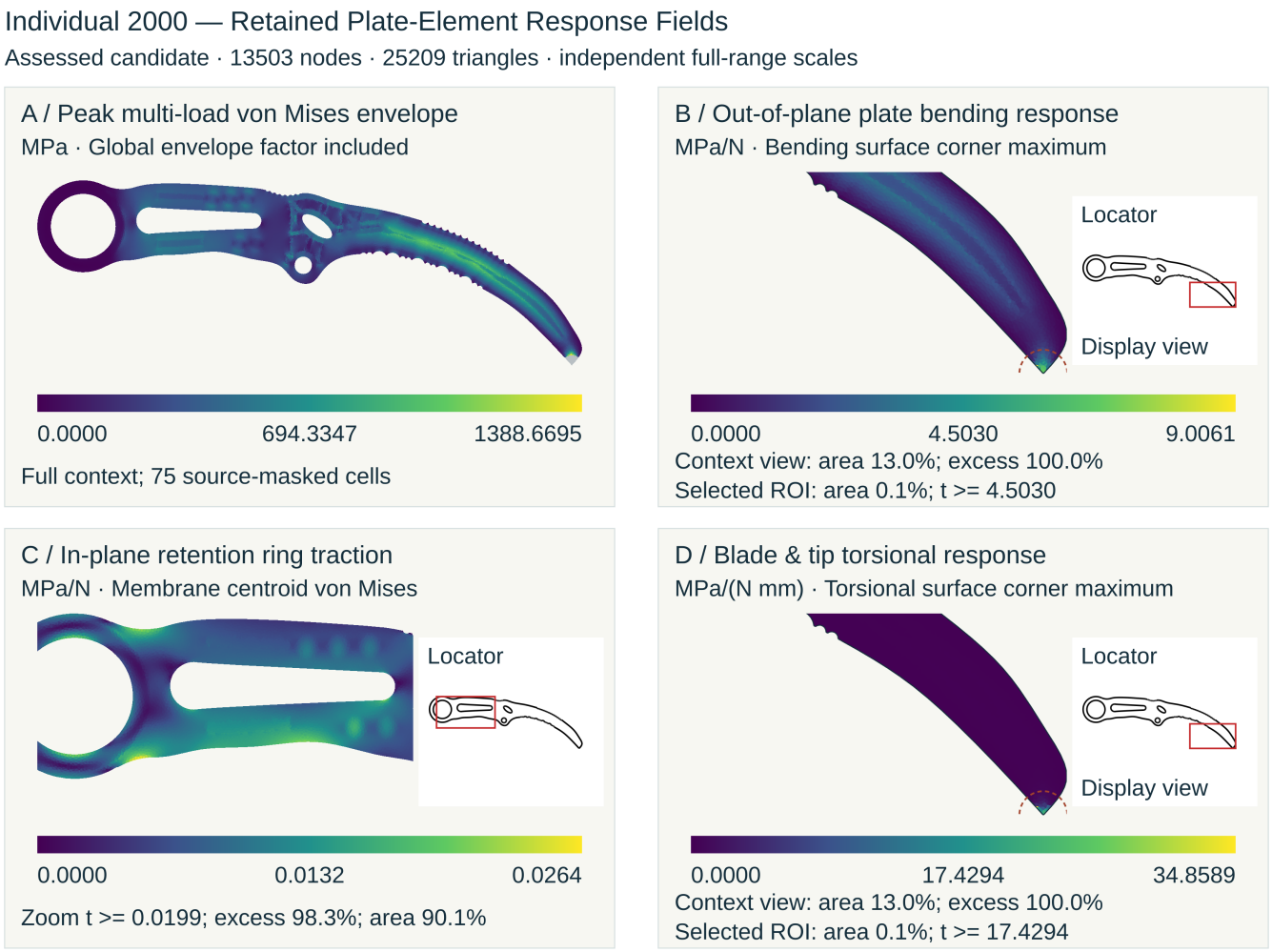


Figure 6 | Source-bound views of Individual 2000's retained plate responses: A, scaled multi-case envelope (MPa); B, unit lateral response (MPa/N); C, unit ring traction (MPa/N); D, raw unit chord torsion (MPa/(N mm)). Each view keeps its full raw color scale, source masks and genuine zeros. B/D widen the camera to show curved blade/tip context; the thin black locator shows the original plate boundary and its red rectangle is the exact displayed viewport. View-area/excess fractions and the smaller selected ROI's area/threshold are reported separately. Coverage uses exact clipped original-triangle areas and each whole eligible field's denominator, not visible ink or local normalization. The source-bound plate-smart-crops.v1.json retains both viewports and statistics. Full context is Supplement Figure 5; in-plane tip pry remains in Supplement Figure 19C.

counts. The opposite tail remains in the full-range overview; its zero-exclusion claim does not apply to the detail selection. Separate per-run point and retained-best context views for Runs 2–11 reuse the same retained births, without connecting different runs into a shared trajectory. Their axes use each run's declared L and key, rather than the pooled overview scale. These additional views add no observations or evidence of physical convergence.

Figure 7 puts every auto-evolve run and evolution wave since 2026-10-17 on the canonical W_ULTRALIGHT_v1: 351 islands and waves on 4 hosts. Each of the 1761 comparable retained Stage-2 assessments (245 from the 2026-10-17/19 waves, 486 from the 2026-10-18 island campaign, 1030 from the continuous evolution running since 2026-10-21) is re-scored from its own scorecard and RS-v2 receipt; stored leaderboard values are not used. Re-assessing retained genomes with the current Stage-2 evaluator reproduces the 2026-10-17/18 scores exactly (80.299390 and 79.869405), so those waves are comparable, whereas an early evaluator (1e93e6f3, 2026-10-16) overstates it: its four re-assessed genomes drop by 11.97–13.12 points (a stored 92.29 re-assesses at 79.166355). Its scorecards and those of a second 2026-10-16 evaluator matched to it by stored metrics, 142 in all, are counted but not plotted. The 2026-10-18 campaign lost the isogrid pockets, lattice and pierced-fuller windows of the 2026-10-17/18 leader (80.299390); the continuous evolution kept them but, until v2.3.1, forced every child's ring wall to at least 6 mm and peaked at 76.799579. After v2.3.1 removed that floor and, in the same release, changed how seeds enter the Hall of Fame (the two changes were not separated), it overtook the leader on 2026-10-23 and

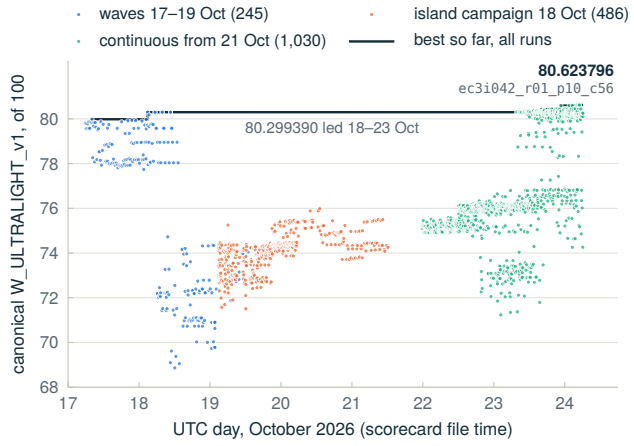


Figure 7 | Every comparable Stage-2 assessment, one canonical scorer: all 1,761 retained Stage-2 assessments of every auto-evolve run and evolution wave since 2026-10-17 (351 islands and waves, 4 hosts), each re-scored with W_ULTRALIGHT_v1 from its own scorecard and RS-v2 receipt and placed at the scorecard's file time. The dark line is the best score so far over all runs, non-decreasing by construction, ending at 80.623796 (`ec3i042_r01_p10_c56`). 142 scorecards from the 2026-10-16 evaluators are not plotted (the re-assessed ones overstate the score). Stage-1 screens are in Supplement Figure 21. Search observations, not physical tests or qualification. Data and census: docs/paper/allruns/.

reached 80.623796 (`ec3i042_r01_p10_c56`, the design on the title page). Supplement Figure 21 summarises 77,915 numeric Stage-1 screens as the best screen of each of 766 island rounds; the earlier waves' screens carry no such value. These are search observations under one scorer, not physical tests, and a cumulative best is monotone by construction. The per-birth view of native Run 5 and its definitions moved to Supplement Figure 12.

9.2 Recorded Ancestry Enables a Later-Assessment Disagreement to Be Inspected

The historical Run 1 retains 1,984 births, recorded within-run parent links, 2176 island-slot memberships and 12 migration events. Parent identifiers for external seed ancestry are not recorded. The genealogy exposes reuse of ancestors rather than a simple branching tree. Its saved FEM-status categories distinguish 1,455 complete records from 529 not evaluated; the latter are not failed solves.

Individual 1095 has the highest stored historical birth score (f = 1.15727, generation 9). Its later strict-assessment summary records cert_pass_all=false and seq_ok=false, although its strict lateral predicate is true. Individual 1543 has the eighth-highest stored birth score (f = 1.15415, generation 13), yet its summary records cert_pass_all=true and seq_ok=true. Thus a lower-ranked birth meets the earlier historical certification predicate that the highest-ranked birth fails. This categorical preference–predicate disagreement is not a continuous later-score rank reversal, and that historical predicate is not the later 26-predicate qualification definition.

These are extrema of retained observations under their native preference directions, not newly evaluated designs. The native maximum has seven ties at serialized precision; its earliest original ID and existing public alias are reported without promoting a qualified champion. The source-bound retained-leader receipt preserves every exact tie and each run's own context. Producer/configuration equivalence across the ten native runs is not admitted, including five runs without retained configuration files. Contextual native fitness, screening deficits, weighted reporting summaries and later assessment scores also have different producer and observation-stage bindings. Their values do not establish a single comparable global best. Missing downstream evidence is not a failed assessment, and missing CAD for a score leader does not make the illustrated example a score winner.

The intermediate campaign retains 3,649 proposals in 57 birth files across 24 run directories. One-based candidate indexing joins 2,259 screens; 1,390 screen observations are missing. A separate retained weighted-reporting subset has six curve files and fifteen generation summaries. In Run 31, the archive score changes from 74.1675 (Individual 2009) to 74.2073 (Individual 2002), then stays at that value. An archive maximum is monotone by construction. The curve alone does not show continuing improvement in physical quality. A compatible versioned weighted recipe is recoverable, but the curve does not establish its exact executed producer or uniform observation stage and is not a controlled convergence experiment.

The automatic leaderboard separately retains one assessed export, Individual 2000, with zero qualified and no promoted champion. The historical 23 × 18 battery includes shared controls, blocked bindings and completeness flags; it is not a set of 414 independent mechanical tests. Individual 2002 also has a retained later-assessment summary: $W_{\mathrm{CAD}} = -7.414831$, $G_1 = 0.259798625$, solid-tip factor 0.3636, 15 battery entries passing and qualified=false. Its underlying exported CAD and full scorecard files are not retained at the referenced paths in the checked repository tree. Its linked proposals and screens are retained in another accession. The earlier screen label "pending independent top-k assessment" is not its final retained assessment-summary status. Neither that summary nor its weighted vector supplies the geometry or response fields of Individual 2000.

9.3 Selected Solver Agreement and Refused Plate Extrapolation Support Different Conclusions

On the same saved solid mesh, the scikit-fem/PyAMG and DOLFINx/PETSc paths agree within 0.05% for the two retained recovered safety-factor observables. Separately, the plate mesh sequence does not admit its selected extrapolated error estimates: one observed order is nonpositive and the other sequence oscillates. Section 11 gives the inputs, recovery definitions and refusal reasons. Close shared-input agreement does not repair the extrapolation assumptions or establish physical qualification.

10 What the Numerical Observations Represent

The retained mechanical observations answer different questions. A unit-load plate field, a later assessment scalar, a shared-mesh solver comparison, and a software-control flag do not provide interchangeable evidence. Table 7 groups the eleven observation families by the conclusion each can support.

Table 6 | Candidate roles and source-bound stored extrema. Each score retains its own profile, stage and source context; the rows are not a common ranking. A public identifier or registry category does not substitute for a retained geometry or assessment artifact.

Individual	Role	Retained evidence	Interpretation
1095 / 1543	Run 1 birth ranks 1 / 8: $f = 1.15727/1.15415$	Birth and earlier certification-summary records	Earlier predicate false / true; not later qualification
781909 (Run 5)	Largest numeric stored native $f = 1.1440$; seven ties	Earliest tied original ID 185541, generation 760; retained population row	No bound per-birth later CAD/solid assessment; qualification unknown
2000	Editorially selected retained assessed example; stable alias, not rank	Exact proposal-content link; bound CAD, plate fields and assessment bundle	Parents unrecorded; zero qualified; no champion
2001 / 2007	Separately assessed controlled exports	CAD/scorecard records; $W_{\mathrm{CAD}} = -3.879/-3.965$ and $G_1 = 0.624/0.622$	Grouped strict verification not met
5443	Smallest retained screen deficit 2.769032384488388; one tie	Source-file ordinal 32, candidate 6; proposal and screen	Ordinal is not generation; full CAD/assessment join unestablished
2002 / 2009	Weighted reporting records in Run 31	Stored $F = 74.207299/74.167488$; linked proposals/screens; 2002 has a later unqualified summary	2002's referenced full CAD/scorecard absent; scorer/stage binding unresolved

The detailed formulas, source keys, and historical catalog are retained in the Supplement; the fixed 23-subject historical battery is separate from the single automatically assessed export, Individual 2000.

10.1 Plate Fields and Assessment Stages

For Individual 2000, the retained plate mesh contains 13,503 nodes and 25,209 triangles. Lateral bending and chord torsion use DKT bending; in-plane tip pry and ring traction use quadratic T6 membrane elements. Their unit-response maps have distinct units and load definitions (Section 11). The scaled multi-case envelope is derived from those maps; it is not an independent fifth load case.

The assessment comparison in Supplement Figure 17C is *certification versus strict*, not a generic coarse/refined pair. Certification takes the minimum capacity over maximum triangle areas 0.25 and 0.125 mm^2, while strict assessment uses 0.0625 mm^2. The saved lateral capacities are 168.90 and 168.18 N; the local tip-torsion factors are 0.524 and 0.529. The displayed fields come from the 0.25 mm^2 level. These stages are distinct from the $4/1/0.25\,\mathrm{mm}^2$ plate mesh study.

The global envelope excludes cells within 4 mm of the tip. The plate body-torsion observable searches beyond 15 mm from the tip; the local tip observable searches the 4–15 mm annulus. Raw torsion imagery retains the tip-patch response and must distinguish its raw peak from the post-exclusion capacity peak. The ring-case producer masks a half-plane beginning 3 mm before the seat; it includes the seat and unloaded blade. Those cells are masked rather than interpreted as measured zero stress.

10.2 Local Solid Response and Recovery

The retained P_2 solid submodel uses 91,008 tetrahedra and 416,583 displacement degrees of freedom. Its cut-arc displacement degrees of freedom are clamped, and vertex forces in a 2 mm tip patch apply 1.82 N m chord torque. The retained comparison is scikit-fem/PyAMG CG versus DOLFINx/PETSc CG/GAMG on that same saved submodel. Rust orchestration is not the primary solid assembly implementation. Section 11 specifies the common inputs, recovery and selected agreement; the publication inputs contain scalar and binned profiles, rather than a full three-dimensional stress field.

10.3 Proxies, Support Assumptions and Missing Observations

The regional yield-order ratio compares modeled elastic capacities and has a stored target of 1.2. It does not model plastic redistribution, cracking or a physical failure sequence. Likewise, the modified-Goodman fatigue factor ($n_f = 0.472$) is a scalar model result, not an observed cycle life.

An alternate support study uses bore-bearing springs and a Winkler grip-pad foundation. Its lateral regional ratio (5.203) remains separate from the certification multi-case ratio (1.435); these differing support/load observables are not a mesh-convergence pair. The original buckling value and pass flag are preserved as historical records, but are withdrawn from interpretation as an admitted buckling-performance result. The producer applies a symmetric eigensolver to $\mathbf{K}_s^{-1}\mathbf{G}$ without a retained operator-admissibility check or eigenpair residual. No replacement eigenvalue or rescored fitness is introduced. Supplement Figure 17A supplies historical support bookkeeping, not a validated stability margin; no support eigenvector field is retained.

10.4 Geometry Records and Control Catalogs

Export identities and reference-contour comparisons bind what was generated and measured. They establish neither physical performance nor human comfort. The historical 23×18 battery and 44-entry crosswalk remain available in the Supplement. They mix candidate-specific observations, shared software controls, missing bindings and completeness flags. In particular, its 20 quadratic-solid completeness labels do not denote 20 favorable strength margins. Catalog definitions, executed controls and unavailable records are reported separately; their counts are not a mechanical replication denominator. The stored plate_gci_valid_3level predicate is historical status bookkeeping: its favorable flag does not establish an admitted extrapolated error estimate when the retained GCI procedure refuses the sequences. Original predicate names and outcomes remain unchanged.

Table 7 | Retained observation families and their limits. The entries describe available records, not newly executed analyses. Fields and assessment scalars shown in the main figures belong to Individual 2000.

Families	Observable	Retained evidence	Supported interpretation
Representation/export; reference shape	B-rep and mesh identity; mass (g); contour distance (mm) and similarity	Bound export records and reference comparison	Inspectable geometry and resemblance to an authored prior; no ergonomic approval
Lateral/pry; ring traction; plate torsion	Unit responses (MPa/N or MPa/(N mm)); capacities (N); regional factors	$0.25\,\mathrm{mm}^2$ cell fields; certification and strict scalars	Elastic responses under named loads and masks; no strict-mesh field is retained
Quadratic solid tip	Recovered vertex stress (MPa) and safety factors	Shared-mesh scalar pairs and distance-bin profiles	Agreement of selected recovered observables; no full solid stress array in the publication inputs
Support/stability; yield-order proxy	Supported regional ratio; unvalidated historical buckling record; elastic capacity ratio	Scalars and a three-level proxy sequence	Dependence on support/region assumptions; no physical fracture sequence
Fatigue	Modified-Goodman factor	Saved scalar $n_f = 0.472$	A model-based cyclic-load margin; no observed fatigue life
Discretization/solver consistency	GCI admission; relative difference of safety factors	Refused plate estimates; solid pairs differing by $< 0.05\%$	Separate evidence about mesh-study admission and shared-input agreement
Software/completeness controls	Contract outcomes, file completeness and score bookkeeping	Historical matrix and control definitions/receipts	Evidence organization and software behavior; no additional mechanical specimens

11 Numerical Agreement and Unresolved Error

11.1 Model, Load and Boundary Identity

The plate and local solid records solve linear-elastic static systems $\mathbf{Ku} = \mathbf{f}$ under their respective geometry, material, support and load assumptions. In the plate path, DKT bending supplies lateral response (MPa/N) and chord-torsion response (MPa/(N mm)); quadratic T6 membrane elements supply in-plane tip-pry and ring-traction response (MPa/N). Lateral and torsion values are per-triangle maxima over three corner-extrapolated bending stresses; pry and ring values are T6 centroid von Mises stresses. The envelope takes the maximum of lateral and pry responses scaled by 260 N, ring traction scaled by 1500 N, and torsion scaled by 1820 N mm with the producer's global torsion correction. That factor is selected at a peak section and applied globally; it is an approximation rather than a locally recovered three-dimensional torsion field.

The local solid model uses quadratic (P_2) displacement fields on linear tetrahedral geometry. It clamps cut-arc vertices and edge-midpoint degrees of freedom to zero. Minimum-norm out-of-plane vertex forces on the 2 mm tip patch supply 1820 N mm chord torque, with zero resultant force and zero transverse moment. The retained producer does not transfer displacements from a larger model. This cut-arc boundary idealization is shared by both solid implementations.

11.2 Field Support and Composite Interpretation

For the 25,209 retained plate triangles, the envelope has 25,134 active cells and 75 producer-excluded cells within 4 mm of the boundary tip node. Its saved active peak is 1388.67 MPa. Ring traction has 10,334 active cells and 14,875 producer-zeroed cells in its source-masked half-plane, which extends beyond the seat. The latter are masked; their zeros are not stress measurements. Lateral, tip-pry and raw torsion maps retain all 25,209 cells, including the tip region that is excluded from selected capacity calculations. Source-coordinate rounding makes one ring-boundary cell ambiguous geometrically; its positive stored ring response establishes that the producer retained it.

The response display averages pooled area-weighted percentile ranks over four available channels, with original source masks and genuine finite zeros kept distinct. The derived envelope receives no additional equal weight, and the available-channel mean has no candidate-specific minimum/maximum stretch. Per-cell contributor records and source-area coverage use four eligible channels as the support denominator; the catalog and metadata inventories are not that denominator. Supplement Figure 3 shows that support. These dimensionless values locate relatively high responses within the retained fields, rather than safety, fitness or quality. Section 8 and the Supplement distinguish the current display definition from its superseded five-map recipe.

11.3 Plate Mesh Study: Refused Error Estimates

The retained three-level study uses maximum triangle areas 4, 1 and $0.25\,\mathrm{mm}^2$, corresponding to the nominal mesh-size convention $h = \sqrt{A_{\max}} = 2$, 1 and 0.5 mm and 3166, 6803 and 25,209 elements. This h is defined from the requested maximum area, rather than a measured element-size distribution. Its stored mesh/observable metadata are valid, but the selected relative-GCI procedure [12] admits neither error estimate:

- The yield-order proxy is monotone across the stored sequence but has observed order $p = -2.9003$; the procedure refuses a nonpositive order.
- The lateral safety-factor sequence has oscillatory differences and is refused on that basis.

The stored extrapolated values and fine-grid GCI estimates remain null. This is a failure to establish the assumptions of this retained procedure for these sequences, not a claim that every numerical-error method is impossible. The certification minimum over $0.25/0.125\,\mathrm{mm}^2$ and strict $0.0625\,\mathrm{mm}^2$ scalars

are separate assessment stages; they do not supply this three-level study with an additional observation.

11.4 Shared-Mesh Solid Agreement

The primary solid path assembles scikit-fem vector P_2 tetrahedra [8] and uses PyAMG smoothed aggregation with CG. The comparison path assembles a DOLFINx 0.9.0 vector Lagrange P_2 space [9] and uses PETSc 3.22.0 CG/GAMG. Both consume the same saved mesh, elastic constants, cut clamp and patch forces. A retained runtime capture lists scikit-fem 12.0.2 and PyAMG 5.3.0; the DOLFINx sidecar records its environment. The capture is written before the orchestration may reuse valid cached solid outputs, and the original solid/verification records do not bind its digest. It therefore documents a captured environment, without proving the exact environment of a cached historical solve. The Supplement gives the full captured versions and image identity separately from the assembly and solver records.

Both recover von Mises stress at element vertices and form the incident-cell maximum or unweighted mean. The comparison region is $4 \le d_{\mathrm{tip}} < 15\,\mathrm{mm}$ with distance from the saved cut vertices $> 3\,\mathrm{mm}$. The saved primary safety factors have three decimal places (0.669, 0.726), whereas the comparison has four (0.6693, 0.7263), under maximum and mean recovery respectively. With $\delta = |\mathrm{SF}_{\mathrm{DOLFINx}}/\mathrm{SF}_{\text{scikit-fem}} - 1|$, those stored rounded scalars give differences of 0.0448% and 0.0413%. These are comparisons of the reported safety-factor tokens, not bounds on unrounded agreement or physical accuracy. The primary record reports 152 iterations and relative residual 8.80×10^{-11}; the comparison reports 158 iterations.

This comparison cross-checks two assembly/solver implementations through selected recovered scalar observables under shared inputs. Common mesh preparation, material, boundary and load assumptions can carry common-mode errors. The comparison does not verify every algebraic operation, bound discretization or modeling error, establish mesh independence, or validate the clamped submodel against physical measurements. The retained profile bins use different plate-cell and recovered-solid-vertex observables and therefore require their own interpretation.

11.5 Open Eigenproblem and Assurance Definitions

The support study's eigenvalue is not admitted here as a current engineering buckling result. Its producer invokes a symmetric eigensolver on $\mathbf{K}_s^{-1}\mathbf{G}$. Symmetric component matrices alone do not establish Euclidean self-adjointness of that product, and no confirming operator-symmetry check or eigenpair residual is retained. The eigensolver's operator contract [11] identifies the unresolved question; it does not supply a historical validation result. The Supplement preserves the original scalar and status as historical provenance. Any qualification inference that depends on this eigenvalue remains unestablished; this reporting revision neither replaces the value nor rescores the historical observations.

The frozen assurance catalog defines 178 challenges, or 356 paired positive/negative control slots across 20 families. Those are definitions, not proof that all slots executed. Individual execution receipts and outcomes must establish which controls ran; unavailable receipts remain unavailable. Neither catalog size nor repeated software-control labels supplies additional candidate-specific mechanical evidence.

12 Reconstruction and Artifact Access

12.1 Four Reproduction Levels

Table 8 defines these four levels and their evidence boundaries.

12.2 Recovery Status

Fresh-process resumption of the native search is unproven in this accession. Serializing a checkpoint or continuing in its original process does not establish disk-only reconstruction of the complete running state in a new process. Separately scoped neutral integer-stream recovery work concerns only its own fixture state, accepted journal prefix and storage protocol; it does not establish resumption of a production campaign. Any recovery claim requires an actual fresh-process receipt binding the fixture source/binary, full restored state, journal sequence and uninterrupted comparison. Such a receipt is distinct from report reconstruction and from a new scientific evaluation.

12.3 Immutable Evidence Entry Points

The retained technical crosswalk provides the public-ID mappings and access to the historical matrix, export identities and original field inventory. The completed-population archive index identifies ten run manifests and 200 compressed segments. These links identify the prior retained evidence accession; they are not a claim that the present revised manuscript was authored at that revision. The final publication receipt supplies the matching manuscript/build accession and standalone supplement access. A TeX include fragment is source, not a compiled supplement download.

Candidate, producer, reporting-source and publication identities remain distinct. For example, a certificate hash binds content identity under its scheme; it does not prove a CAD artifact exists. The archived producer revision is preserved with its original inputs rather than replaced by the present manuscript revision. The matching publication receipt must bind the actual manuscript head/tree, build environment, main and supplement PDF hashes, generated outputs and receiving-main/release accession. Receiving-tree equality establishes the delivered source bytes; it does not reconstruct an unrecorded historical producer execution. Hash equality establishes byte identity, not correctness of the original computation.

12.4 Reporting Commands and Writes

The reporting environment requires Rust/Cargo with the locked dependencies and complete admitted archives; document production additionally requires LuaLaTeX/IEEEtran, the declared fonts, librsvg and PDF inspection tools. From the repository root, the following commands request a full compressed-population audit and retain the existing document-build/check entry points. Commands are shown at full width so flags remain copyable; actual exits/toolchain/output hashes belong to the final validation receipt.

The artifact audit decodes admitted historical frames, checks proposal/screen joins, verifies compressed-segment and reporting bindings, and checks generated-output consistency. Its flag-controlled compressed-segment decoder defaults to one

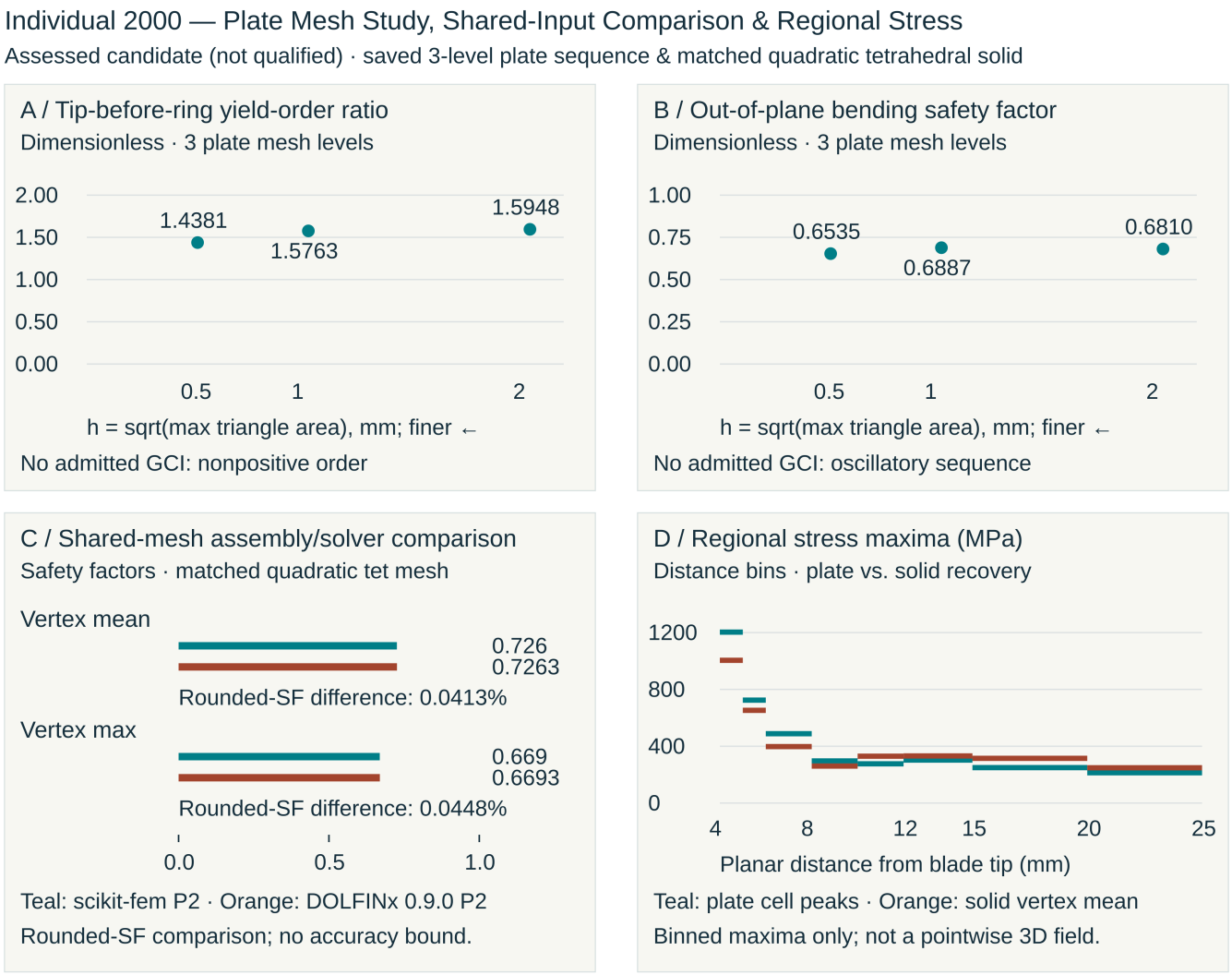


Figure 8 | Individual 2000 retained numerical observations. A–B: plate sequences with maximum triangle areas $4/1/0.25\,\mathrm{mm}^2$; the stored relative-GCI estimates are refused for nonpositive order and oscillatory differences. C: scikit-fem/PyAMG and DOLFINx/PETSc safety factors on the same $91{,}008$-tetrahedron, $416{,}583$-degree-of-freedom solid submodel, using incident-cell vertex maximum and unweighted mean recovery; primary safety factors are stored to three decimal places and comparison values to four. D: distance-bin stress maxima for raw plate cells and recovered solid vertex means; the differently recovered profiles are not a pointwise field-error estimate.

Table 8 | Distinct reproduction claims and their scope in this study. The publication/release receipt identifies actual command outcomes for its exact source and outputs.

Level	Operation	Evidence boundary
Report reconstruction	Read retained contracts/archives; check identities; regenerate reporting views and documents	The publication operation; preserves evidence inputs but writes derived/build outputs
Solver replay	Rerun a bound CAD/FEM workflow under a declared environment	Not performed by the reporting rebuild
Software controls	Execute named positive/negative fixtures and admission checks	Actual receipts establish executed cases; catalog definitions alone do not
Physical replication	Fabricate specimens and obtain mechanical observations	Physical observations are not reported in this study

segment per run; --full-zstd removes that limit. For the admitted completed-population accession, full scope is 200 segments containing 1,954,560 rows. The final exact-head receipt must establish the requested mode, actual segment/row counts and successful exit; documenting the command alone does not establish its execution. The document check renders figures

and compiles the paper under target/ieee/paper/, and the existing build also updates docs/paper/main.pdf. Evidence inputs remain read only; the entire operation is not filesystem read only. The supplement target, complete input manifest, all-page visual inspection and clickable-link checks are part of the matching publication package rather than implied by the existence of source files.

13 Prospective Representation Research

FunSearch provides context for separating language-model program proposals from execution-based evaluation in mathematical and algorithmic settings [6]. It does not establish an agent result in Jarambit. Searching values in the present schema and proposing a different schema or decoder are distinct activities; the retained runs demonstrate the former.

A future controlled study on benign reference CAD tasks could compare human-authored and agent-assisted representation proposals under fixed evaluator versions and review criteria. A proposal would identify types, units, dependencies, serialization and migration rules, and expected tests; retained rejected proposals would permit measurement of invalidity, reconstruction success, and reviewer effort. Those outcomes remain unmeasured here. Human responsibility for the problem definition and interpretation remains essential, while independently controlled evaluators constrain which proposed effects are actually observed.

14 Discussion

14.1 What the Connections Establish

The case study makes three relationships inspectable. Ancestry records connect a birth to recorded sources of variation; candidate and stage identities connect a search preference to a particular assessment; source artifacts connect displayed quantities to the models and transformations that produced them. The usefulness lies in being able to follow these relationships for a concrete result, rather than inferring them from a final geometry or score.

The historical comparison illustrates why the assessment predicate must travel with an outcome. A lower-ranked birth can meet the run's recorded predicate when the highest-ranked birth does not; that observation is meaningful within its stated selection and assessment context. The numerical comparison has an analogous structure: agreement on a shared solid discretization supports consistency of selected recovered observables, while the plate refinement study addresses admission of a different error estimate. Preserving both observations makes their respective implications visible.

14.2 Scope and Limits

This is a retrospective study of one repository and selected retained workflows. Prior selection and retention determine which observations are available, and missingness is not assumed random. Births share ancestry and search context; numerical observations can share geometry, preprocessing and model assumptions. Counts therefore describe record coverage, rather than independent experimental replication. Contextual fitness and cumulative best-so-far curves describe the recorded preference; neither those curves nor selected assessment comparisons estimate a population-wide qualification rate or establish comparative search advantage.

The mechanical observations are computational model outputs. Shared-input solver agreement leaves discretization, loading and boundary idealizations to their own checks; the support-study buckling estimate is not admitted because its operator/eigenpair assumptions remain unverified. Physical observations are not reported. Material labels and CAD dimensions also leave manufacturing state and process capability to measurement, as illustrated by NIST's test-artifact work [10]. A visible motif and an authored feature family establish neither its autonomous invention nor an isolated engineering benefit. Those questions require additional evidence.

14.3 A Testable Research Direction

The next question is whether explicit links improve reconstruction and interpretation on benign scientific workflows. Controlled provenance defects could test detection and explanation, and independent readers could measure reconstruction accuracy, time and uncertainty. Such comparisons would evaluate the usefulness of the reporting approach itself. The present case supplies a concrete artifact and specific questions for that evaluation.

15 Conclusion

This study followed one computational design result from its representation to its evidence. A versioned representation and decoder, recorded evolutionary search and a declared multi-objective score produced `ec3i042_r01_p10_c56`, the highest of 1,761 comparable Stage-2 assessments at 80.623796 of 100, and a complete re-assessment gave the same value. Surface-tension pocket smoothing then replaced its chipped pocket walls with clean contours inside the CAD export, and the full assessment of the new solid, not its appearance, gave 80.561227. At every step the retained links say which stage, which scorer and which export a number belongs to.

Three lessons come from the records rather than from the design. *A ranking is a declared preference.* W_ULTRALIGHT_v1 ranks every candidate continuously, and the design on the first page passes 14 of 18 battery evaluators and is not qualified. *A comparison is only as good as its scorer.* In the four re-assessed cases, stored values from an early evaluator exceeded the canonical score of the same genome by 11.97–13.12 points; only recomputation from retained scorecards made the census comparable. *A change of geometry is a new design.* One smoothing setting raised the score of the design it was chosen on and lowered that of the current best, so it is assessed design by design.

What the study does not establish is equally specific. It reports no physical observation, no qualification, no comparison with other optimizers or with expert designs, and no isolated benefit of any feature. Agreement between two solvers on a shared mesh checks implementation, not the model, and the plate mesh study refuses its extrapolated error estimates.

The arrangement is not specific to a knife. The search consumes scores; the representation, the decoder and the tests carry the domain. Where a design can be written in an executable representation and its consequential behaviour

Table 9 | Prospective application classes: what a search would vary and what evidence each domain would require before a computed preference could become an accepted part. Rows are prospective; none is a result of this study.

Domain	What a search varies	Evidence required before use
Aerospace and spaceflight fixtures	bracket topology, sections, fastener layout	validated FE; coupon-to-component tests; vibration qualification
Medical-device research	lattice, section, porosity	standard endurance tests; biocompatibility; clinical evidence for the intended use
Robotics	finger shape, compliance, flexures	grasp trials on standard object sets; fatigue of flexures
Thermal management	channels, fins, lattices	conjugate models checked against instrumented test loops
Scientific instruments	mounts, kinematic couplings	dimensional metrology; thermal-drift and stiffness measurement
Energy systems	flow paths, supports	operating-envelope tests; degradation and cycling data

can be measured by tests that remain credible where a search will go, the same loop can propose and rank candidates for fixtures in aerospace and spaceflight hardware, components for medical-device research, grippers and compliant parts in robotics, passages for thermal management, mounts for scientific instruments and parts of energy systems. Each domain adds its own obligations before a computed preference can become an accepted part: validated models, dimensional metrology and process characterization for the manufacturing route, and physical tests at the conditions of use (Table 9). Additive manufacturing widens the set of geometries that can be made; it does not by itself supply precision, fatigue performance or qualification.

The claim, stated conditionally

Given a representation that can express useful designs, an objective that says what is preferred, and tests that are credible where the search goes, computational search can explore designs and trade-offs that are impractical to enumerate by hand, and it can leave an evidence trail that another researcher can follow. This paper demonstrates that trail for one computational case. Proof for any built object remains specific to that object and its intended use.

Data and Code Availability

All source, retained evidence, receipts and renders are public at github.com/neverhuman/jarambit. Every generated number in the paper comes from crates/paper-artifacts and a named receipt; the Supplement maps each figure and table to its inputs and hashes. Rebuilding the paper reconstructs reports only; it runs no new search, CAD export or solve.

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